

Benchmarking of NQCH's quantum computer

March 16, 2026

1. Report of Changes

Platform: Unknown
Calibration-id: N/A
Calibration date: Unknown
Calibration note: N/A

Experiment-id: N/A
Experiment date: Unknown
Experiment note: Unknown

Platform: Unknown
Calibration-id: N/A
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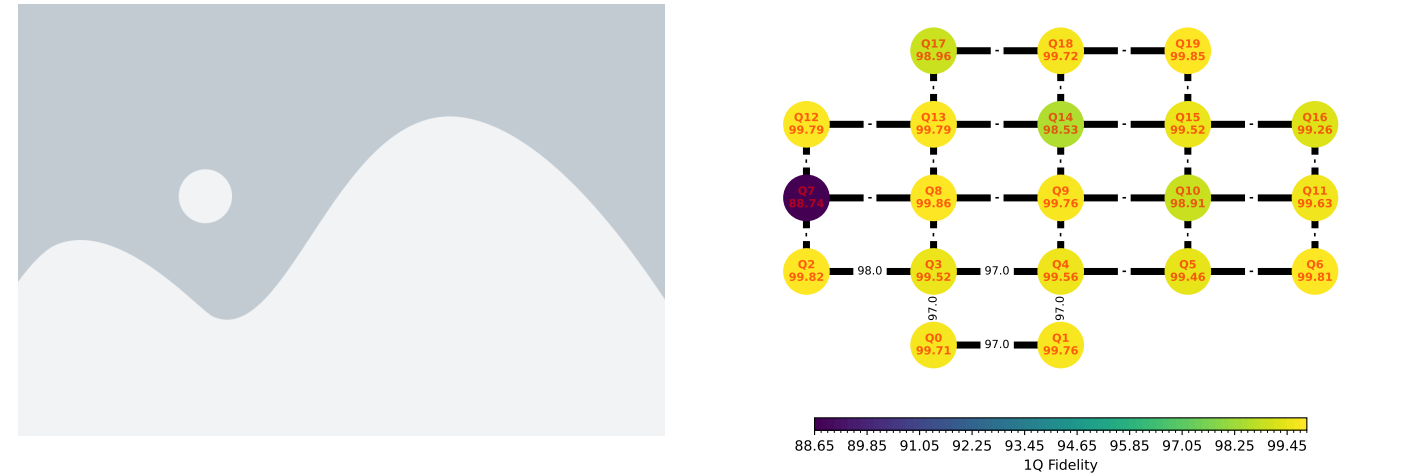
2. Version Comparison

Library	Version	Library	Version

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3. One and two qubit fidelities

The single qubit fidelity is obtained via Randomized-Benchmarking. The two-qubit fidelity is the "Bell-state fidelity".



4. Statistics

	Average	Median	Min	Max
T1 (ns)	-	-	-	-
T2 (ns)	-	-	-	-
Fidelity	-	-	-	-
RO Fidelity	-	-	-	-
2q RB Fidelity	-	-	-	-
2q CZ Fidelity	-	-	-	-

	Average	Median	Min	Max
T1 (ns)	1.28e+04	1.23e+04	646	3.65e+04
T2 (ns)	2.36e+25	3.68e+03	125	9.43e+26
Fidelity	0.99	0.997	0.887	0.999
RO Fidelity	0.794	0.777	0.777	0.927
2q RB Fidelity	0.965	0.972	0.923	0.981
2q CZ Fidelity	0.975	0.974	0.967	0.991

5. Best Qubits Selection

k-qubits	Best Qubits	Fidelity
2	N/A	N/A
3	N/A	N/A
4	N/A	N/A
5	N/A	N/A

k-qubits	Best Qubits	Fidelity
2	2, 3	0.981
3	0, 2, 3	0.976
4	0, 1, 2, 3	0.970
5	0, 1, 2, 3, 4	0.965

6. Summary metrics

Benchmark	Metric	Value	Runtime
Mermin	Max	-	-

Benchmark	Metric	Value	Runtime
Mermin	Max	-3.316	1.02 sec

7. Randomized Benchmarking Results

Qubit n	Fidelity	Error Bars	Qubit n	Fidelity	Error Bars
0	0.997	± 0.000131	10	0.989	± 0.000582
1	0.998	$\pm 7.47e-05$	11	0.996	± 0.000218
2	0.998	$\pm 7.57e-05$	12	0.998	± 0.000126
3	0.995	± 0.000163	13	0.998	$\pm 7.15e-05$
4	0.996	± 0.000185	14	0.985	± 0.00183
5	0.995	± 0.000147	15	0.995	± 0.000469
6	0.998	$\pm 8.34e-05$	16	0.993	± 0.000558
7	0.887	± 0.0287	17	0.99	± 0.000502
8	0.999	$\pm 4.48e-05$	18	0.997	± 0.00026
9	0.998	$\pm 7.2e-05$	19	0.998	$\pm 9.05e-05$

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8. Mermin

Runtime	Qubits	Max
-	-	-

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1.02 sec	0, 2, 3	-3.316

